Project Proposal

1. Project Summary

The present study investigates the acoustic characteristics of singing modes using Machine Learning and **Digital Signal Processing (DSP)** techniques. The singing modes considered in the study are **Chest**, **Head**, **Mask**, and **Breathy** modes. The study aims to quantify these singing modes using Machine Learning and DSP techniques. The engineering approach includes source separation of commercial audio and feature extraction of the singing modes. The relationship between the fundamental **frequency (F0)** and **harmonics (Hn)** will be used to develop a classifier for identifying singing modes in various music genres. The study aims to develop a system that classifies singing modes and also statistically analyzes the distribution of timbre in various music genres.

2. Broader Impacts

This research bridges the gap between musicology and data science. For **Vocal Pedagogy**, it provides an objective feedback loop for students to visualize resonance ("Mask") and register changes (Chest vs. Head) through data rather than intuition. The study is also highly beneficial for **Music Information Retrieval (MIR),** it offers a new way of tagging songs based on the 'mood' of the singer. The study also offers a new approach to **Speech Pathology** because it can be used to quantify 'breathiness' in singing. 'Breathiness' is a critical factor in speech and singing. The study also offers a new approach to analyzing the evolution of singing modes in the digital music age.

3. Data Sources

- **VocalSet:** Wilkins, J., Seetharaman, P., Wahl, A., & Pardo, B. (2018). *VocalSet: A Singing Voice Dataset for the Deep Learning Age.* [https://zenodo.org/record/1193957]
- **GTSinger:** Wang, J., et al. (2022). *GTSinger: A High-Quality Chinese Singing Voice Dataset with Technique Labels.* [https://github.com/vocal-dataset/GTSinger]
- **Spleeter (Tool for Isolation):** Hennequin, R., Khlif, A., Felix, J., & Monte, M. (2020). *Spleeter: A Fast and Efficient Source Separation Information Library.*

4. Expected Major Findings / Project Questions

- **Hypothesis 1: Chest voice** will exhibit a significantly lower **Spectral Slope** (stronger high-frequency harmonics) compared to **Head voice**, even when normalized for volume and pitch.
- **Hypothesis 2: "Mask" resonance** can be objectively identified by a concentrated energy peak (Formant) between **2.5 kHz and 3.5 kHz**, regardless of the vowel being sung.
- **Question 1:** To what extent does the **Harmonic-to-Noise Ratio (HNR)** decrease during "Breathy" phonation across different vocal ranges?
- **Question 2:** Is there a statistically significant correlation between specific genres (e.g., R&B vs. Rock) and the use of non-harmonic techniques like **Vocal Fry**?
- **Outcome:** A visualization of the **"Timbral Fingerprint"** of different techniques, showing how energy shifts from F0 to higher harmonics as a singer moves from Head to Chest voice.